Shortest path

The shortest path algorithm

Michel Bierlaire

Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

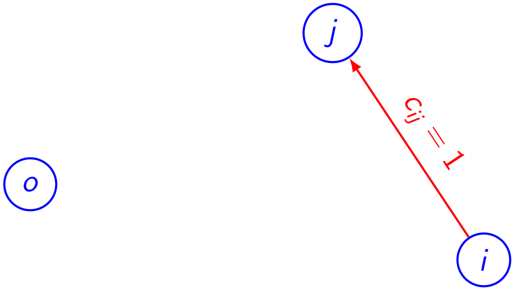
The shortest path algorithm

We consider here the single origin–all destinations shortest path problem
The optimality conditions are based on labels associated with each node
The idea of the algorithm is to make sire that the labels verify the optimality conditions.
The label of node j can be interpreted as the length of the shortest path identified so far from the origin to node j.

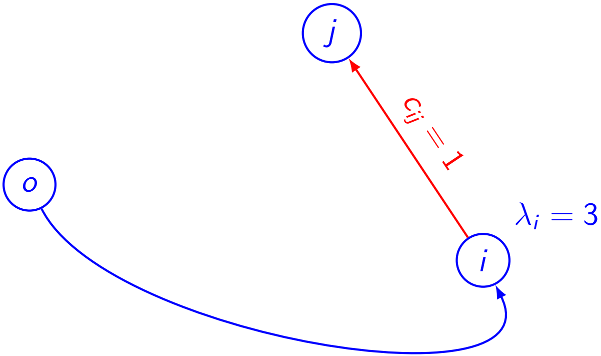
Optimality conditions

$$\lambda_j \leq \lambda_i + c_{ij}, \quad \forall (i,j) \in \mathcal{A}.$$
$$\lambda_j = \lambda_i + c_{ij}, \quad \forall (i,j) \in P.$$

Labels



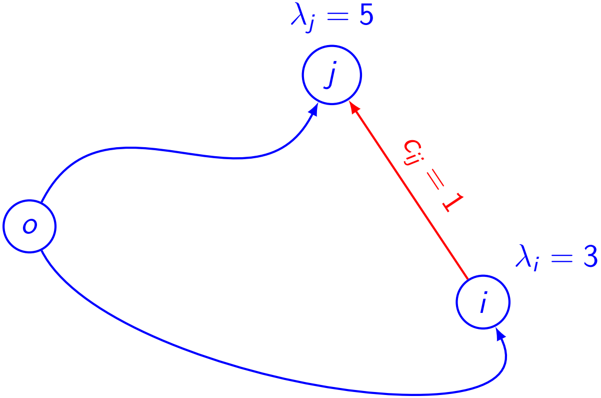
Labels



Labels

Write that $\lambda_j > \lambda_i + c_{ij}$
Update the label from 5 to 4 by hand.

Labels



Idea of the algorithm

For each arc (i, j) in the network, if

$$\lambda_j > \lambda_i + c_{ij},$$

$$\lambda_j \leftarrow \lambda_i + c_{ij}.$$

Second idea

Loop on the m nodes instead of the n arcs.

$\mathcal{S} \leftarrow \{o\}$, $\lambda_o = 0$, $\lambda_j = +\infty$.

- Select i in $\mathcal{S}$.
- For each (i, j):
  - If $\lambda_j > \lambda_i + c_{ij}$,
    - $\lambda_j \leftarrow \lambda_i + c_{ij}$.
    - If $\lambda_j < 0$ and $\lambda_j < (m-1)\min_{(i,j)\in\mathcal{A}} c_{ij}$: STOP.
    - $\mathcal{S} \leftarrow \mathcal{S} \cup \{j\}$.
- $\mathcal{S} \leftarrow \mathcal{S} \setminus \{i\}$.
- If $\mathcal{S} = \emptyset$: STOP.
- Otherwise, start again.

Summary

The shortest path algorithm updates labels at each iterations.
It maintains a list of nodes to be treated.
Treating a node amounts to consider each outgoing arc, and updates the labels if needed.
If a label is updated, the corresponding node is added to the set of nodes.
The algorithm stops when all nodes have been treated, or if the problem is unbounded.